

Lecture10: Classification2

Logistic Regression

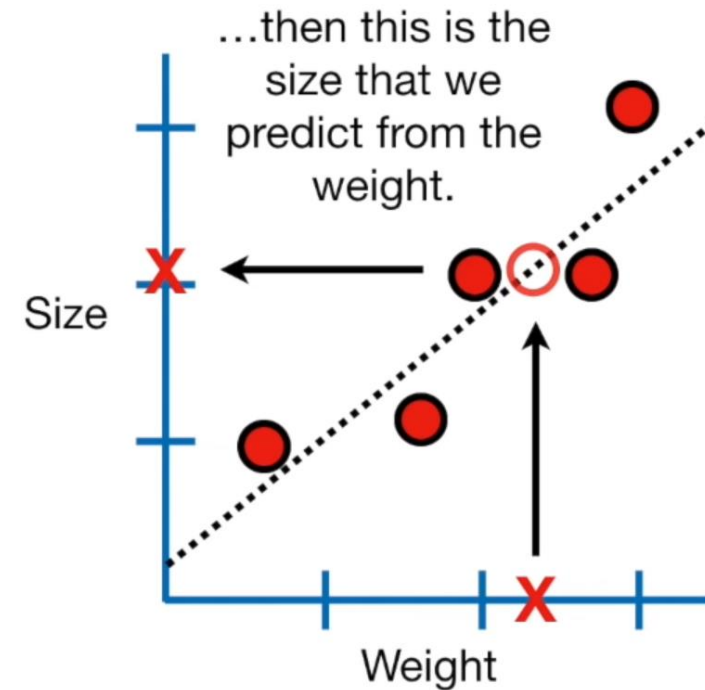
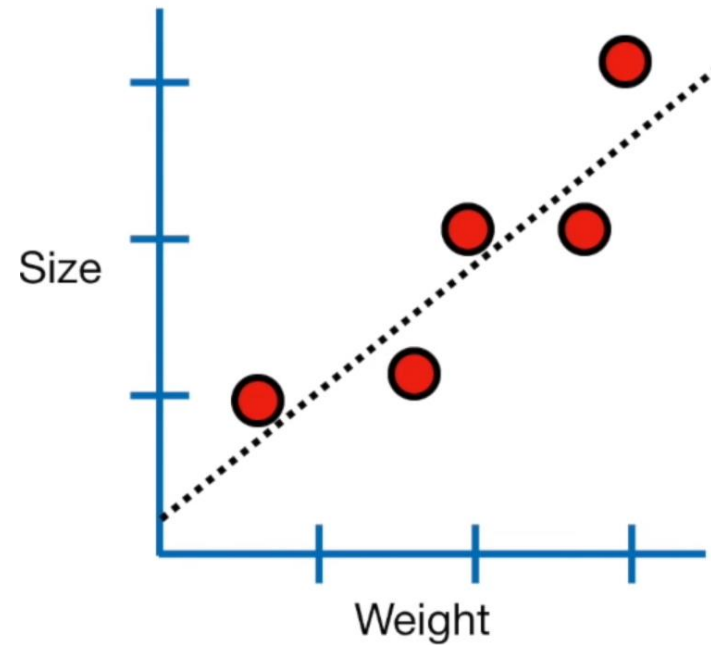
STA 4724

HanQin Cai

University of Central Florida

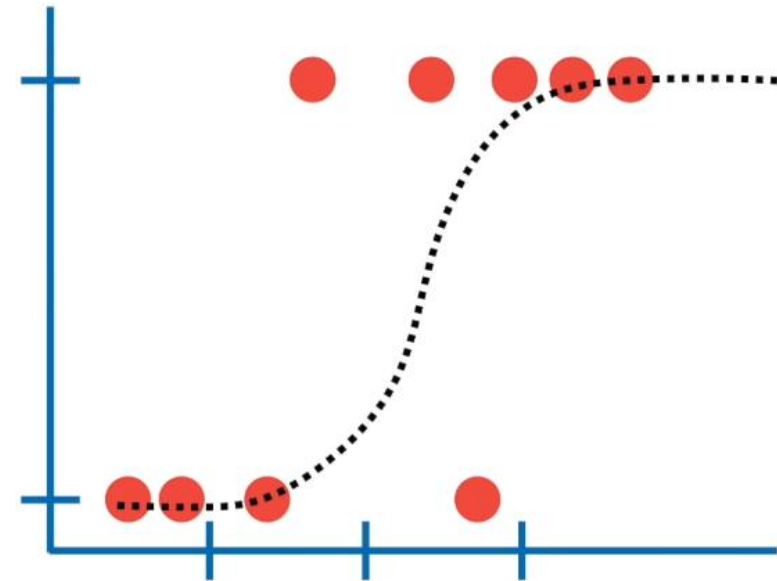
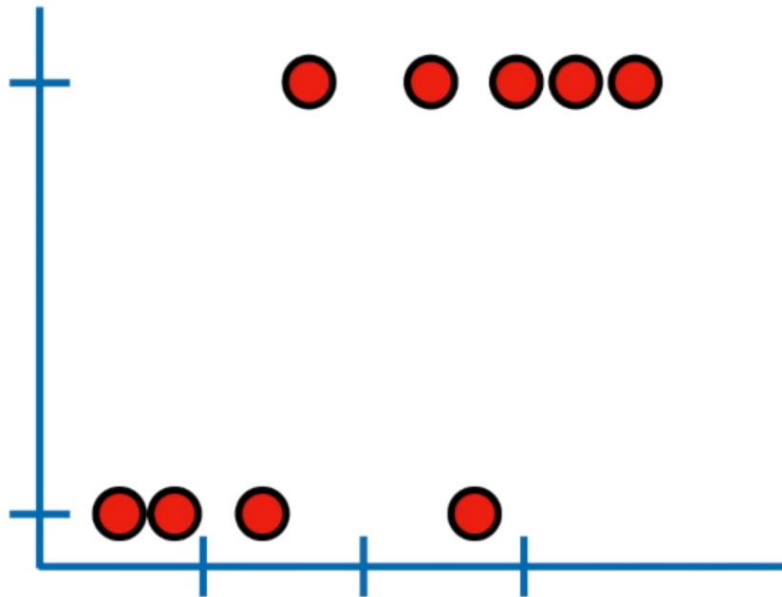
Linear Regression

- Recall what we learned few weeks ago
- Can be used to predict a continuous numerical value



In Classification Problem

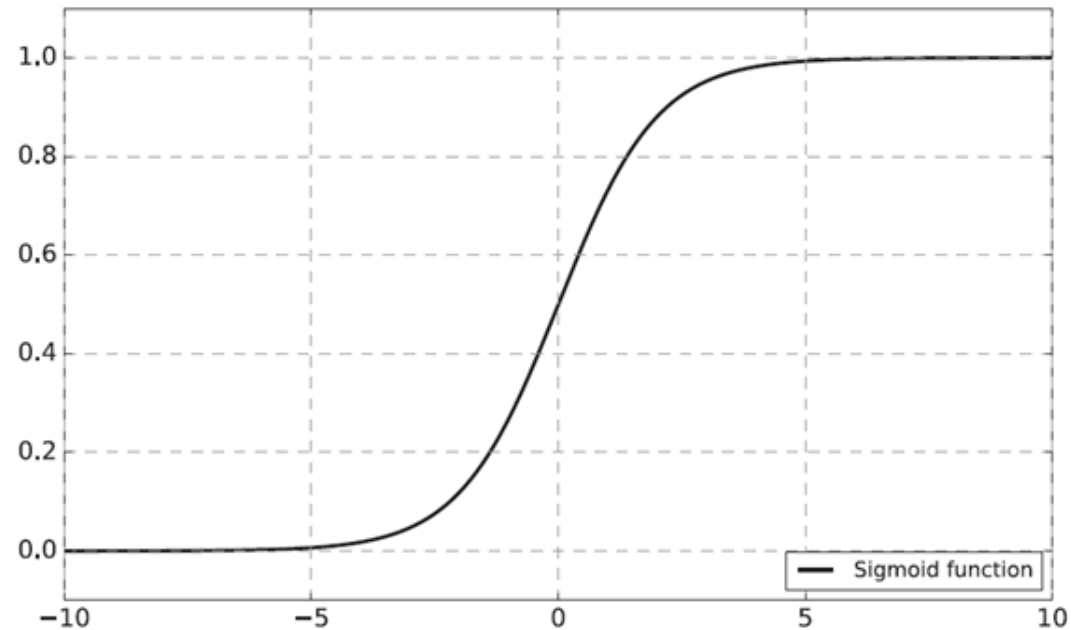
- We are looking for discrete output (e.g., True(1) or False(0))
- For a smooth fitting, we seek the S shape curve.



Logistic function

$$g(z) = \frac{e^z}{1 + e^z}$$

- $g \in [0,1]$ for all $z \in (-\infty, \infty)$
- Also known as sigmoid function – a widely used activation function in NN



Logistic Regression

- In linear regression, $Y = X\beta + \varepsilon$ with ε being zero-mean.
$$\mathbb{E}(Y|X) = X\beta$$

- In logistic regression, we want

$$\mathbb{E}(Y|X) = \frac{\exp(X\beta)}{1 + \exp(X\beta)}$$

- Since the result is binary ($Y = 1$ if X is true, $Y = 0$ otherwise), so
$$\mathbb{E}(Y|X) = \mathbb{P}(X)$$

Logit function

- For a probability p , the **odds** is defined by

$$\frac{p}{1-p}$$

- The **logit** function or **log-odds** function is

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

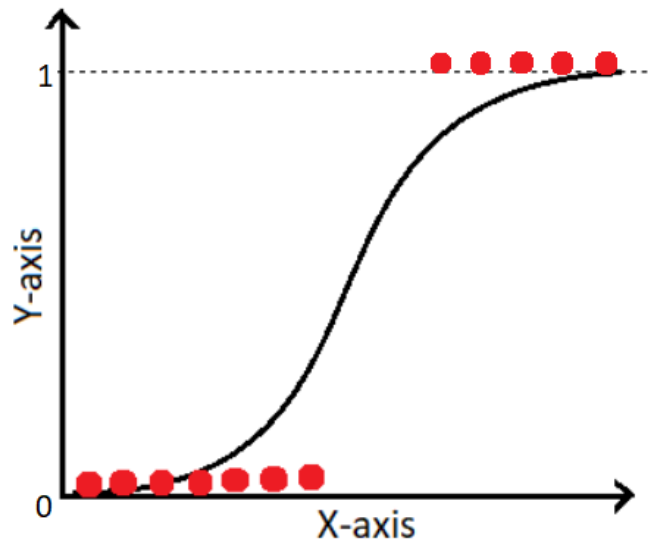
Do some math

$$\begin{aligned} & \log\left(\frac{\mathbb{P}(X)}{1 - \mathbb{P}(X)}\right) \\ &= \log \mathbb{P}(X) - \log(1 - \mathbb{P}(X)) \\ &= \log\left(\frac{\exp(X\beta)}{1 + \exp(X\beta)}\right) - \log\left(1 - \frac{\exp(X\beta)}{1 + \exp(X\beta)}\right) \\ &= \log(\exp(X\beta)) - \log(1 + \exp(X\beta)) + \log(1 + \exp(X\beta)) \\ &= X\beta \end{aligned}$$

With logit function, we can recover the linear model from the logistic model

Maximum Likelihood Estimation (MLE)

- Logistic regression model as a generalized linear model
- You may have heard people talking about the equivalence between MLE and ordinary least square
- The close form solution of ordinary least square is not the recommended solver
 - least square implicitly assume the noise is zero-mean normally distributed



Maximum Likelihood Estimation (MLE)

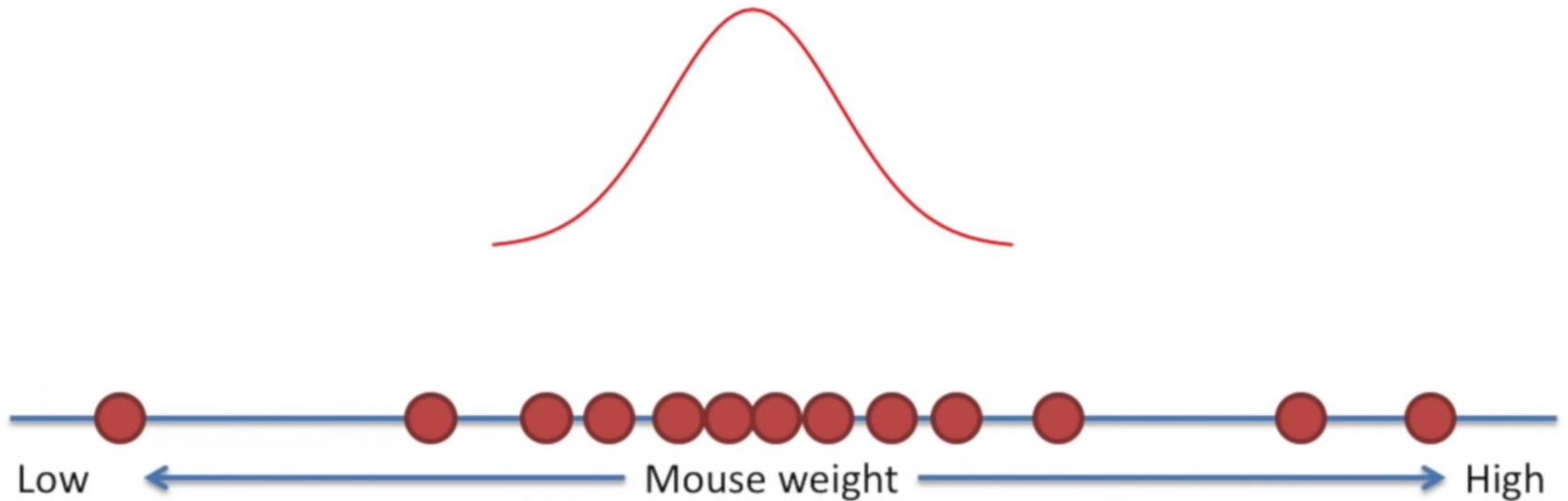
- We want to maximize the log-likelihood of observing the training data

$$\beta = \max_{\beta} \sum_i \log \mathbb{P}(Y_i; X_i, \beta)$$

- You can also add regularization to the objective function, for example

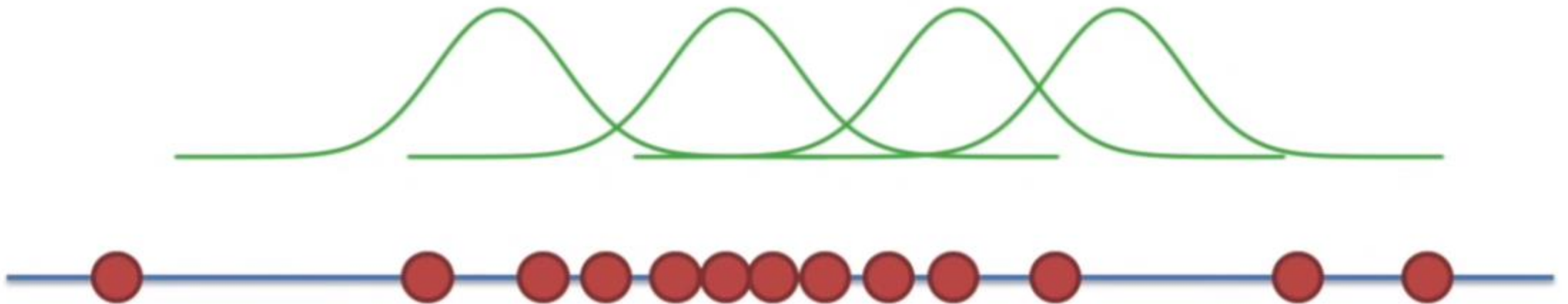
$$\beta = \max_{\beta} \sum_i \log \mathbb{P}(Y_i; X_i, \beta) - \lambda ||\beta||_1$$

Maximum Likelihood Estimation (MLE)

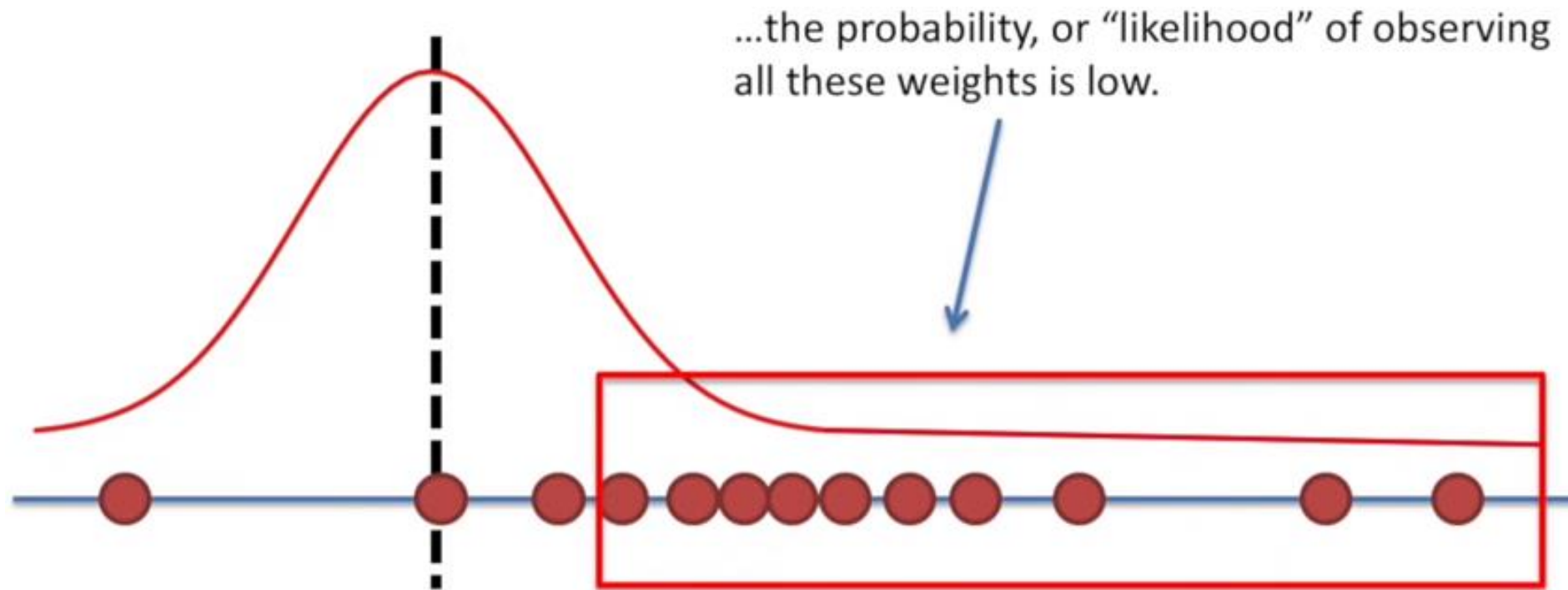


Maximum Likelihood Estimation (MLE)

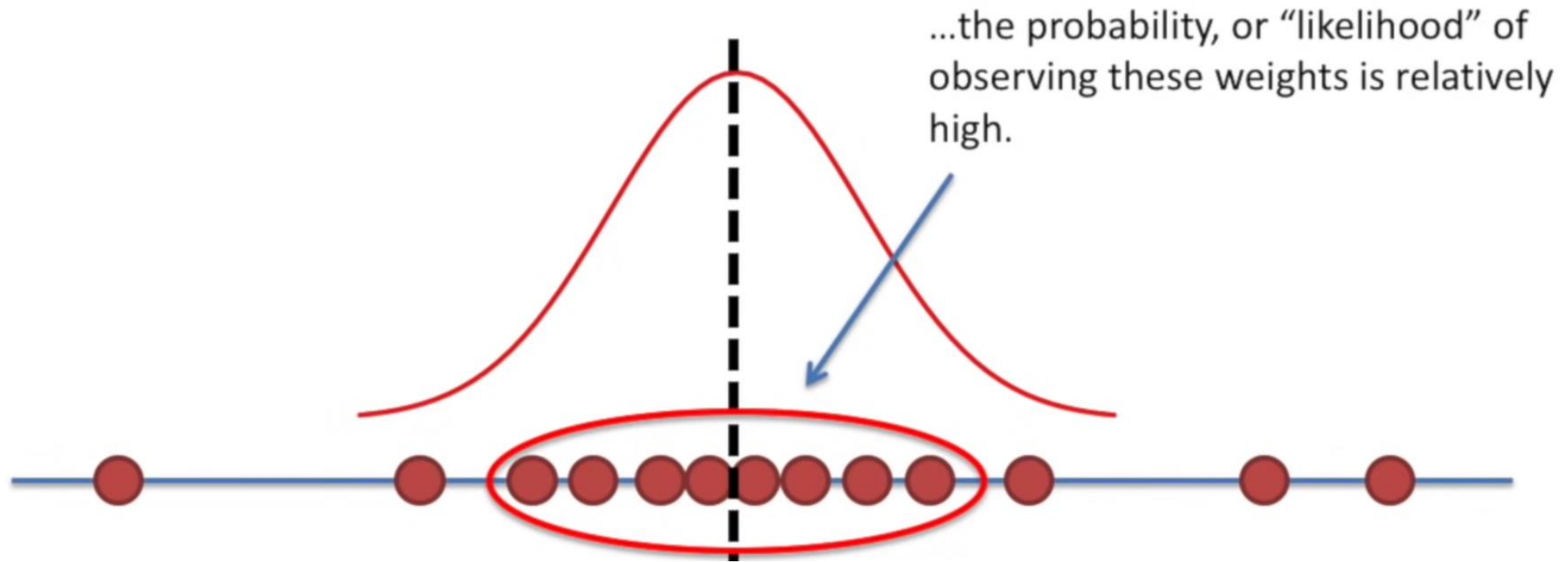
Is one location “better” than another?



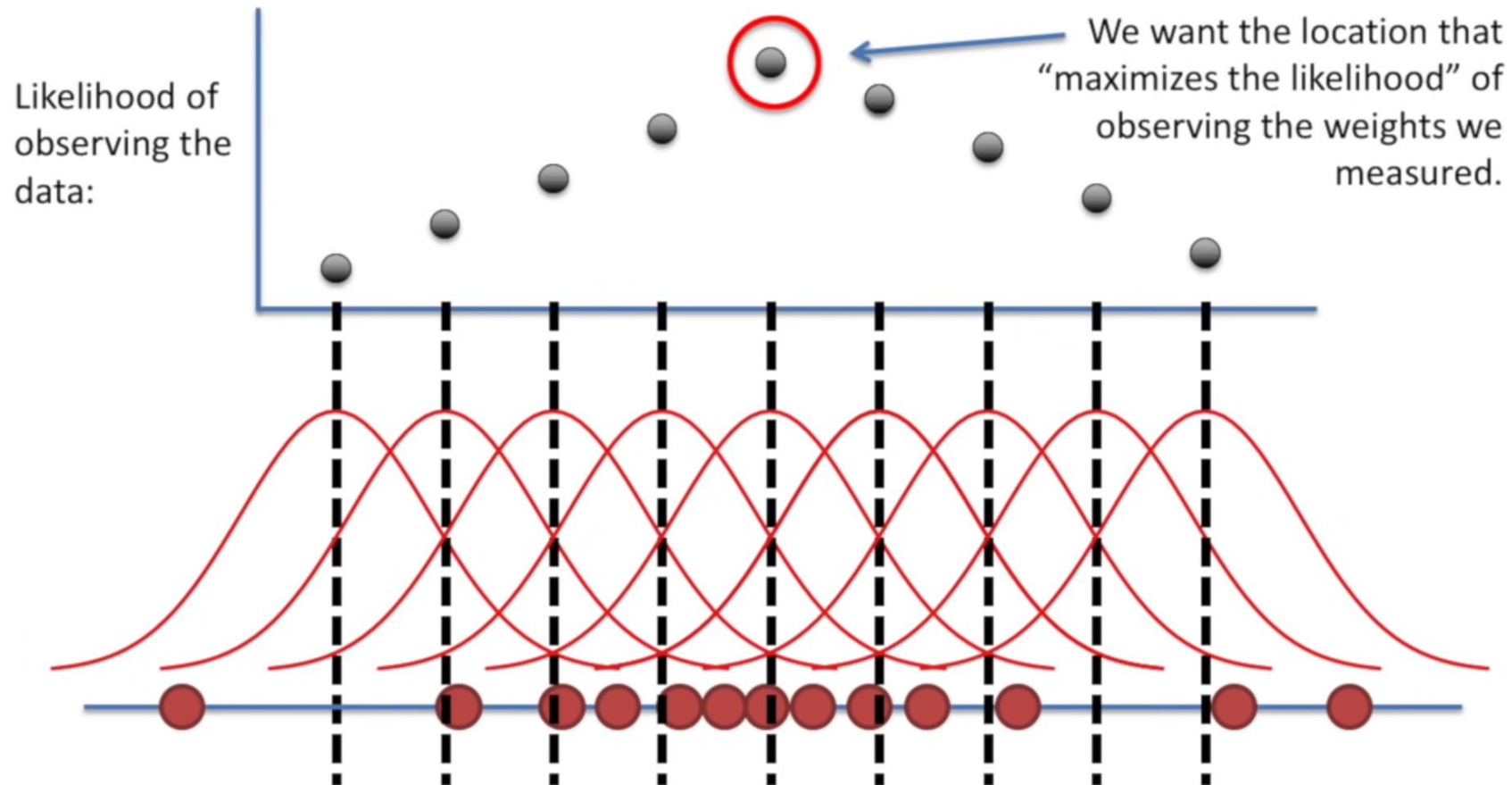
Maximum Likelihood Estimation (MLE)



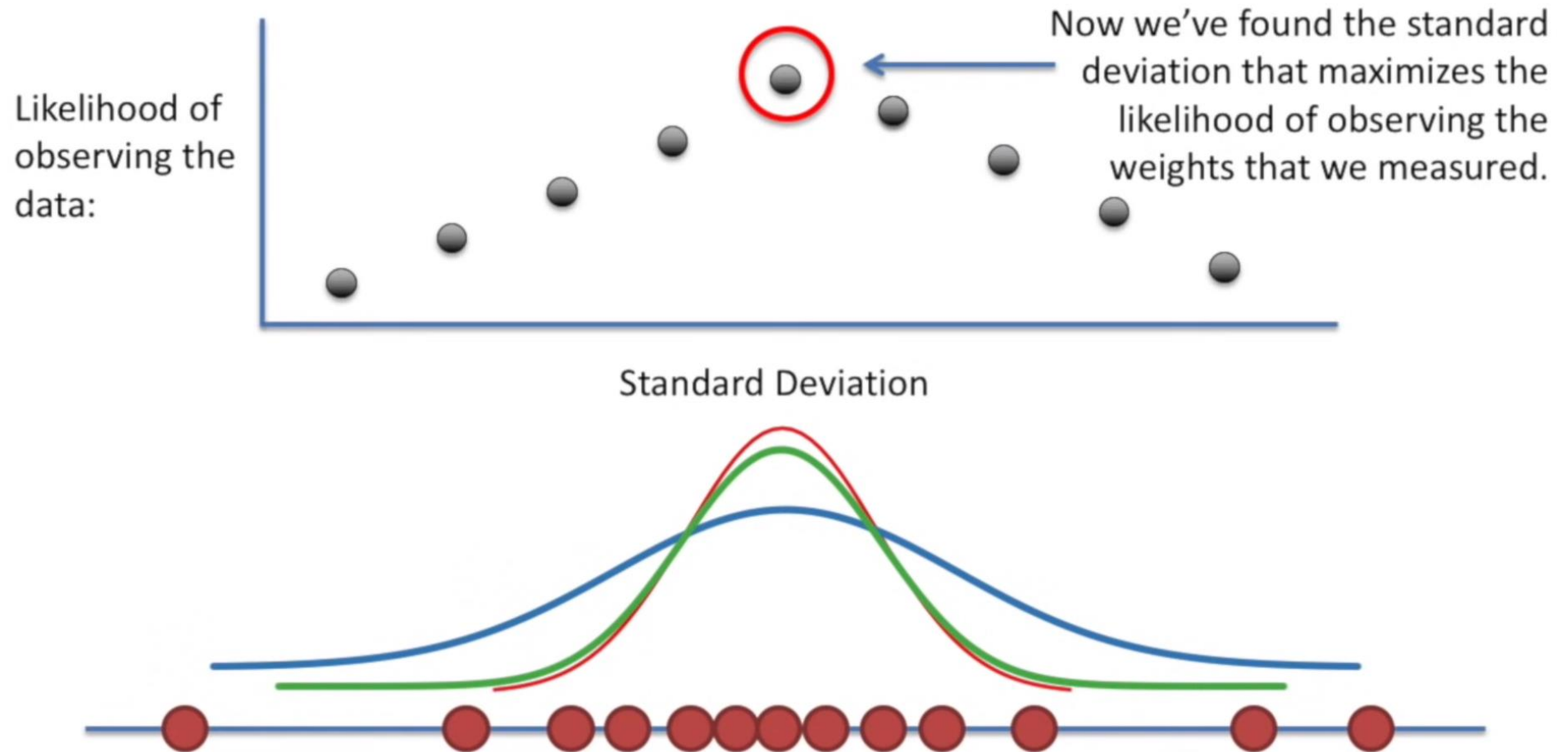
Maximum Likelihood Estimation (MLE)



Maximum Likelihood Estimation (MLE)



Maximum Likelihood Estimation (MLE)



Threshold

- The output of a logistic model will provide us the probability that an observation belongs to a binary class or not
- In the final step of logistic regression, we employ a threshold to carry out the classification result

$$Y_i = \begin{cases} \text{True (1)} & \text{if } \mathbb{P}(X) > 0.5 \\ \text{False (0)} & \text{otherwise} \end{cases}$$

Multi-Class Regression

- Logistic regression can be used in a multiclass setting too and a typical strategy is the so-called *one-versus-the-rest* strategy
- For example, if we have a problem with three classes 0, 1 and 2, we will need three classifiers:
 1. Classifier one for class 0 versus 1 and 2
 2. Classifier two for class 1 versus 0 and 2
 3. Classifier three for class 2 versus 0 and 1

Time to Play

- Let's see how to use Logistic Regression in **python**!

